EXP 16

```
clc;
clear;
close all;

% Simulation and Analysis of Uniform and Non-Uniform Quantization

%% 1. System Parameters
Fs = 10000;             % Sampling frequency
t = 0:1/Fs:0.05;        % Time vector
fm = 100;               % Message frequency (Hz)
A = 1;                  % Signal amplitude
nBits = 3;              % Number of quantization bits
L = 2^nBits;            % Number of quantization levels

%% 2. Generate Message Signal
x = A*sin(2*pi*fm*t);

%% 3. Uniform Quantization
xmin = -A;
xmax = A;

delta = (xmax-xmin)/L;

% Mid-rise uniform quantizer
indexU = floor((x-xmin)/delta);
indexU = min(max(indexU,0),L-1);

xU = xmin + (indexU+0.5)*delta;

%% 4. Non-Uniform Quantization
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% Mu-law compression, uniform quantization, expansion
mu = 255;

% Compress the normalized input signal
xComp = sign(x).*log(1+mu*abs(x)/A)/log(1+mu);

% Uniformly quantize the compressed signal
deltaC = 2/L;
indexN = floor((xComp+1)/deltaC);
indexN = min(max(indexN,0),L-1);

xCompQ = -1 + (indexN+0.5)*deltaC;

% Expand the quantized signal
xN = sign(xCompQ).*A.* ...
   ((1+mu).^abs(xCompQ)-1)/mu;

%% 5. Calculate Quantization Errors
errorU = x-xU;
errorN = x-xN;

%% 6. Calculate MSE
MSE_U = mean(errorU.^2);
MSE_N = mean(errorN.^2);

%% 7. Calculate SQNR
signalPower = mean(x.^2);

SQNR_U = 10*log10(signalPower/MSE_U);
SQNR_N = 10*log10(signalPower/MSE_N);
```

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%% 8. Plot Original and Quantized Signals
figure('Name','Quantization Analysis');

subplot(3,1,1);
plot(t,x,'b','LineWidth',1.3);
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
stairs(t,xU,'r','LineWidth',1.1);
hold on;
plot(t,x,'b--');
title('Uniform Quantization');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Quantized','Original');
grid on;

subplot(3,1,3);
stairs(t,xN,'m','LineWidth',1.1);
hold on;
plot(t,x,'b--');
title('Non-Uniform Quantization (Mu-law)');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Quantized','Original');
grid on;

%% 9. Plot Quantization Errors
```

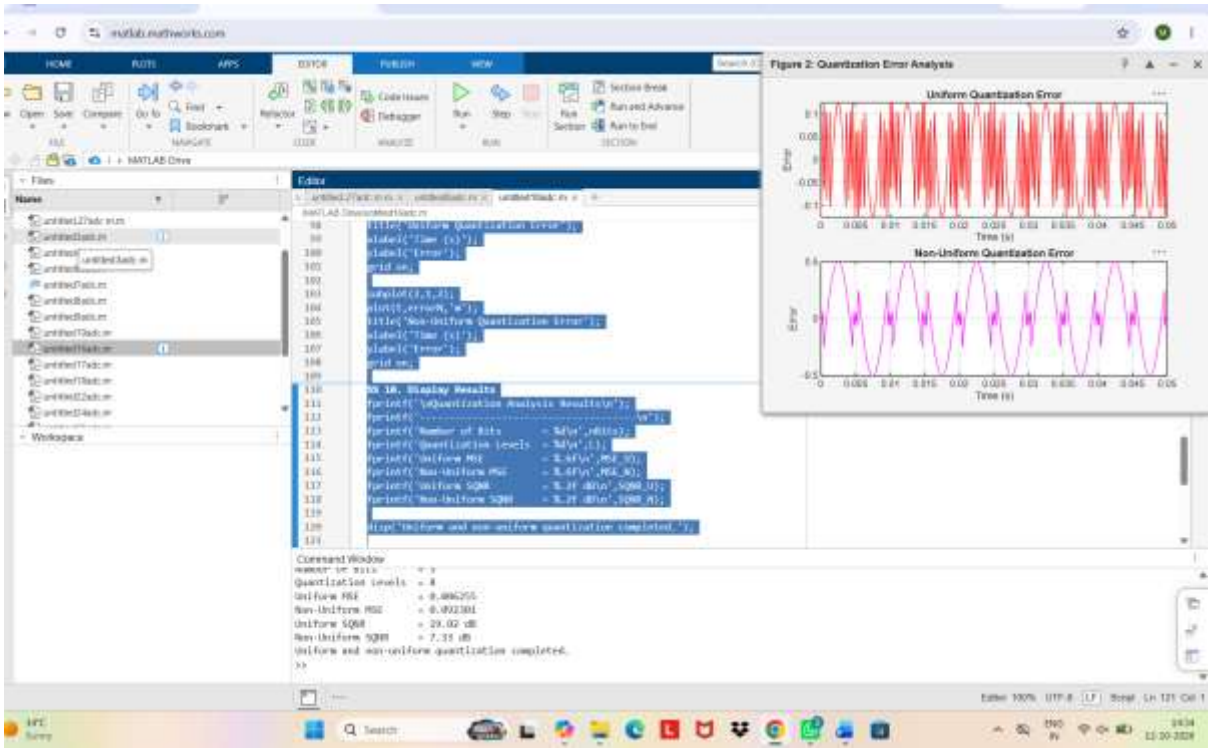
```
figure('Name','Quantization Error Analysis');

subplot(2,1,1);
plot(t,errorU,'r');
title('Uniform Quantization Error');
xlabel('Time (s)');
ylabel('Error');
grid on;

subplot(2,1,2);
plot(t,errorN,'m');
title('Non-Uniform Quantization Error');
xlabel('Time (s)');
ylabel('Error');
grid on;

%% 10. Display Results
fprintf('\nQuantization Analysis Results\n');
fprintf('--------------------------------------\n');
fprintf('Number of Bits       = %d\n',nBits);
fprintf('Quantization Levels  = %d\n',L);
fprintf('Uniform MSE          = %.6f\n',MSE_U);
fprintf('Non-Uniform MSE      = %.6f\n',MSE_N);
fprintf('Uniform SQNR         = %.2f dB\n',SQNR_U);
fprintf('Non-Uniform SQNR     = %.2f dB\n',SQNR_N);

disp('Uniform and non-uniform quantization completed.');
```


Figure 2: Quantization Error Analysis
Uniform Quantization Error
Error
Time (s)
Non-Uniform Quantization Error
Error
Time (s)
Command Window
Quantization levels = 8
Uniform MSE = 0.006255
Non-Uniform MSE = 0.002381
Uniform SQNR = 19.02 dB
Non-Uniform SQNR = 7.33 dB
Uniform and non-uniform quantization completed.